MOD300 Anvendt Python programmering og modellering

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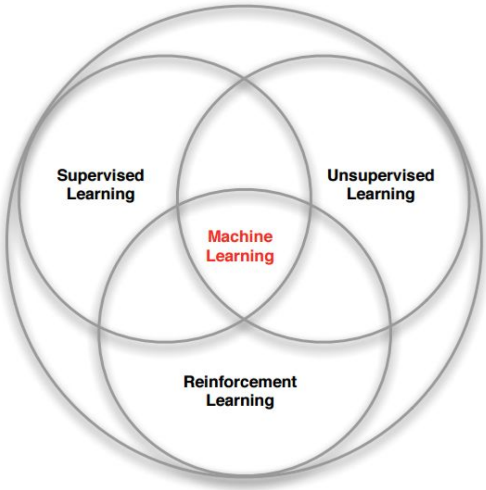
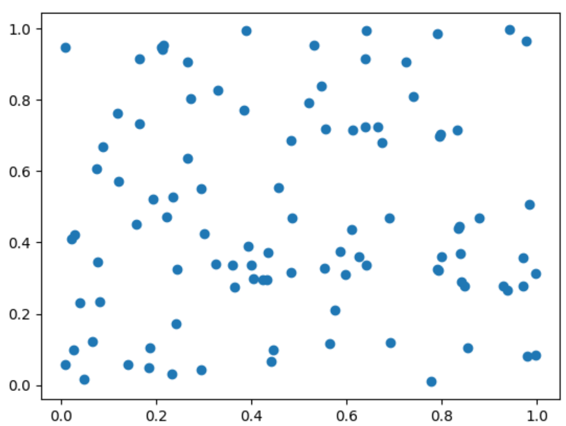
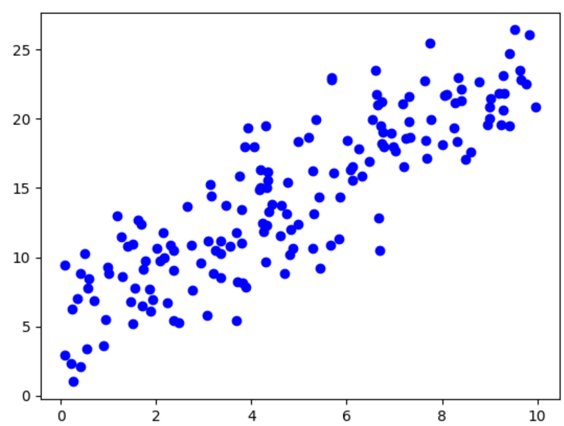
Families of Machine learning



What can we do with that?



What about in this case?



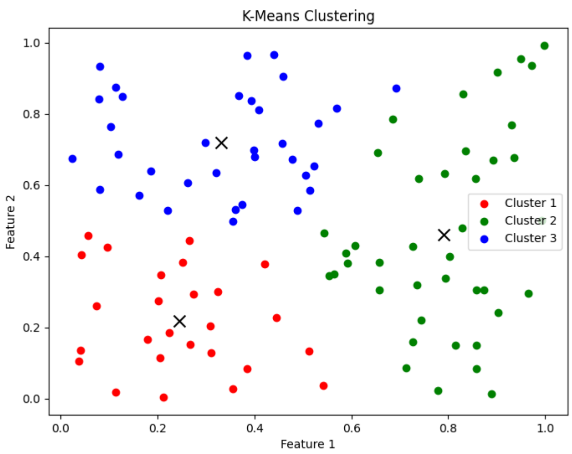
Python Source code 1

```
import numpy as np
import matplotlib.pyplot as plt

# Generate some sample data
data = np.random.rand(100, 2)   # 100 data points with 2 features

plt.scatter(data[:, 0], data[:, 1])
plt.show()
```

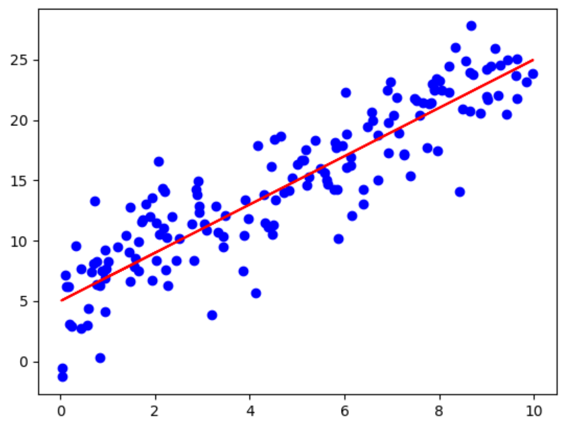
Python Source code 2

```
import numpy as np
import matplotlib.pyplot as plt

def generate_linear_data(n_random_points, noise=16):
    x = np.random.rand(n_random_points) * 10

    # Make 'perfect' data
    true_slope,  true_intercept = 2, 5
    y = true_slope * x + true_intercept

    # Add noise
    y += np.random.randn(n_random_points)*noise

    return x, y, true_slope, true_intercept

# Use the function to generate data
x, y, true_slope, true_intercept = generate_linear_data(
        n_random_points=166,
        noise=3)

# Plot all
plt.scatter(x, y, color='blue', label='Data Points')
plt.show()
```

Unsupervised learning



Supervised learning



The data decides

This is why we focus so much on the data type.

The data properties dictate what statistical model can be adopted.

An statistical model has leverages our understanding of the data structure to improve its **predictions** (inference).

The numerical recipe that we used to generate the data is defined the **truth**

Psychology or data science?

Most Machine learning tools are aimed to find the truth. In most cases, we are happy to not find lies.

Unsupervised learning

Unsupervised learning, a term that resonates with the autonomy of machine intelligence, operates on the principle of identifying patterns and structures in datasets without labelled responses.

This branch of machine learning is distinguished by its lack of explicit guidance, where algorithms are tasked with uncovering hidden structures from unlabeled data.

The most common clustering strategies are :

- filtering
- clustering
- dimensionality reduction
- association learning

Application of unsupervised learning

It is a bit of a holy grail: a computer that finds patterns without guidance. (Yes, it doesn't work, most of the time)

Still, it has been shown efficient for:

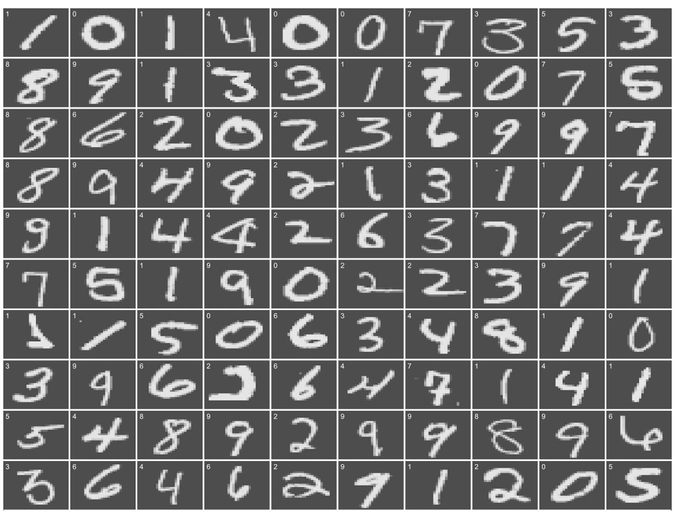
- Computer vision
- Anomaly detection
- Exploratory data analysis

Main challenge

The right result is quite undefined, Uncertain goal.

We will demonstrate it with a famous problem.

Uncertain goal



Weak supervised learning

A less popular type of machine learning problem is when labels are assigned to groups of instances.

The group of instances is called **bag**.

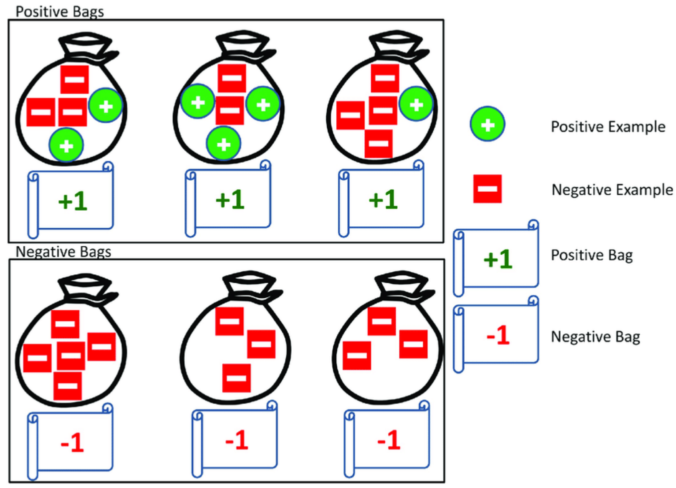
The question is, what is the level of a previously unforeseen bag?

This data structure and question type request a hybrid treatment between supervised and supervised learning.

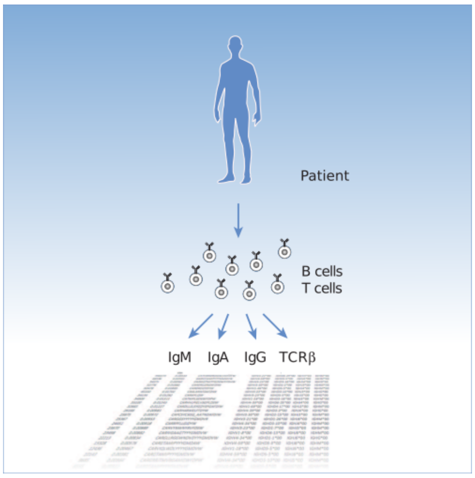
Multiple instance learing

Multiple instances are needed to learn (quite clear name)

Weak Supervised learning



Weak Supervised learning



Reinforcement learning

Finally, there is a further approach.

Reinforcement learning (RL)

It aims to train an intelligent agent to take actions in a dynamic environment in order to maximise the cumulative reward.

It learns from outcomes and decides which action to take next. After each action, the algorithm receives feedback that helps it determine whether the choice it made was correct, neutral or incorrect.

It is a self-teaching system that essentially learns by trial and error.

It is a dependable tool for automated decision making.

Classification

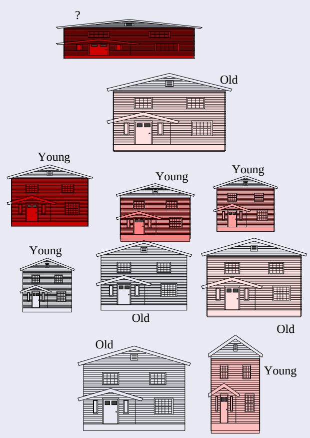
- UnSupervised classification is about "non-guided" **discrimination**
- Supervised classification is about "guided" **discrimination**
- Eeach object/sample has **external information** (or class membership)

Aim:

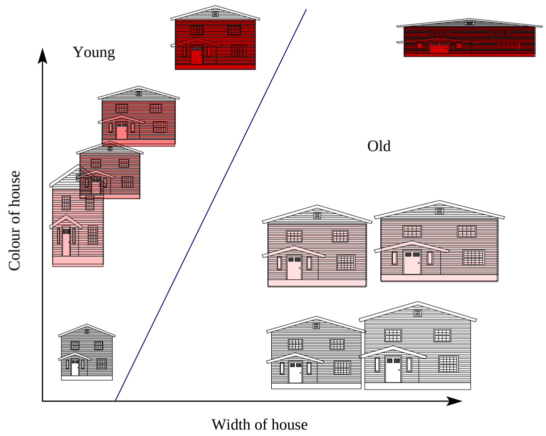
- To find a **rule** for the membership of new objects
- Identify **what attributes** are important

Classification

Do we see any rule?

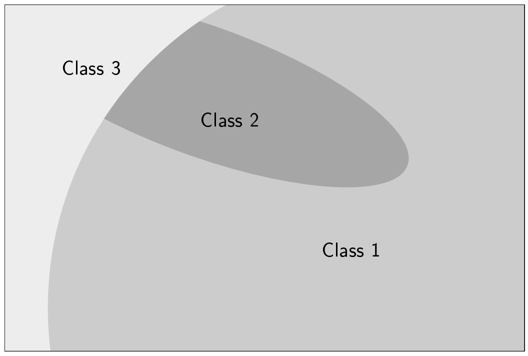


Partitioning the attribute space



Decision boundaries

Many classification methods can be viewed as a **partitioning** of the feature space such that each partition of space represents a **class**. Partitioning performed by **decision boundaries** (can be linear or curved)



Discriminant functions

Probability functions behave like a **classifier** which assign a class to a new sample object: $\mathbf{x}$ to class c_i if:

$$g_i(x) > g_j(x), \quad \forall i \neq j$$

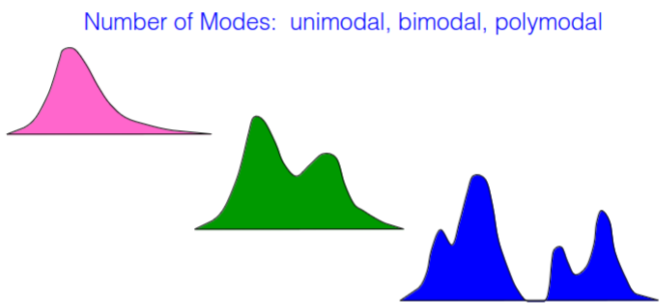
The **decision surfaces** resulting from using discriminant functions are defined as:

$$g_{ij}(\mathbf{x}) \equiv g_i(\mathbf{x}) - g_j(\mathbf{x}) = 0, i \neq j.$$

Probability distribution

A metric is needed

Amount of uncertainty



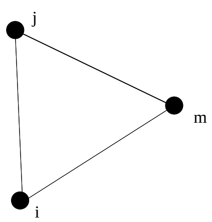
Metrics

Triangle inequality

Considering a vectors in an N-dimensional space, to be a distance it must satisfy the triangle inequality:

$$d_{ij} + d_{im} \geq d_{jm}$$

If also $d_{jj} = 0$, if $d_{ij} - d_{ji} = 0$, then we call it a *metric*.



Metris example

Common metrics:

- Counting
- Euclidean

$$d_{ij}^{(E)} = \left[\sum_{k=1}^{N} (x_{ik} - x_{jk})^2 \right]^{\frac{1}{2}}$$

Categorical variables do not necessarily satisfy these relations.